THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which observation shows that a plant is making more food than it uses? 1

(a) it turns pale **(b)** it stops growing

(c) it loses mass **(d)** it gains mass and grows

2. A 20 cm object is placed 40 cm from a lamp and its shadow falls on a screen at 100 cm from the lamp. The shadow height is: 1

(a) 40 cm **(b)** 20 cm **(c)** 50 cm **(d)** 100 cm

3. During a lunar eclipse the Moon takes 1 hour to cross the Earth's umbra. The umbra's width relative to the Moon's diameter is about: 1

(a) about 2.5 times **(b)** 10 times **(c)** half **(d)** equal

4. Bread mould is a: 1

(a) autotroph **(b)** insectivore **(c)** saprotroph **(d)** parasite

5. Which of these helps us to see a non-luminous book? 1

(a) the shadow of the book

(b) light from a lamp reflecting from the book

(c) the book giving its own light

(d) light passing through the book

6. A student thinks 'the dark side of the Moon is always dark'. This is: 1

(a) correct on full moon only **(b)** correct

(c) incorrect **(d)** correct on new moon only

7. Which of the following does not increase the rate of transpiration? 1

(a) high temperature **(b)** wind **(c)** high humidity **(d)** dry air

8. A girl stands 2 m from a mirror and a wall is 5 m behind her. The image appears: 1

(a) 2 m behind the mirror **(b)** 7 m behind

(c) on the wall **(d)** 5 m behind the mirror

9. If the Moon revolved around the Earth in the same plane as the Earth's orbit, eclipses would occur: 1

(a) randomly **(b)** every month **(c)** only once a year **(d)** never

10. A plant in deep shade may not grow since: 1

(a) photosynthesis is less than respiration **(b)** it gets more light
(c) roots are weak **(d)** it gets more CO2

11. A stick 1 m high casts a shadow of 0.8 m. A tower casts a shadow of 40 m at the same time. The height of the tower is: 1

(a) 50 m **(b)** 32 m **(c)** 40 m **(d)** 80 m

12. A shadow stick has a shadow of 0 length at noon on the equator on 21 March. This shows: 1

(a) no light **(b)** the Sun is at horizon
(c) it is night **(d)** the Sun is overhead

13. **Assertion (A):** Transpiration is the loss of water vapour from the leaves through stomata. 1
Reason (R): Transpiration takes place only through the roots of a plant.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** A plane mirror forms a virtual image. 1
Reason (R): The image formed by a plane mirror can be obtained on a screen.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The Moon is a natural satellite of the Earth. 1
Reason (R): The Moon does not produce its own light.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A plane mirror forms an image of a 5 cm tall object. What are the image height and nature? 2

17. Why is the Pole star not exactly at the geographic north? 2

18. Why is alcohol used to remove chlorophyll in the starch test? 2

19. Explain how a pinhole camera could be used to observe a solar eclipse safely. 2

20. Why does the shadow of a vertical stick trace a curve through the day? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why does the Sun rise earlier in eastern India than in western India? 3

22. Explain how leaf shape and size differ in different habitats and why. 3

23. Explain why a mirror placed in a dark room cannot be seen. 3

24. Explain what happens to day and night at the equator through the year. 3

OR

Why are greenhouse owners increasing carbon dioxide levels and using lamps? 3

25. A 6 cm candle is placed 30 cm from a pinhole of a camera whose length is 15 cm. Find the image height. 3

OR

A shadow stick 1 m long casts a shadow of 0.5 m at noon in summer and 1.5 m in winter. Compare the Sun's height and explain. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Prakash lives in Delhi. He notices that in June the days are very long and hot, while in December the days are short and cold. His cousin in Sydney, Australia, says that it is cold there in June. 5

(i) Name the main causes of seasons.
(ii) What season is it in Delhi in June?
(iii) On about which dates are day and night nearly equal?
(iv) Why is it winter in Sydney when it is summer in Delhi?

27. Explain how a plane mirror in a barber's shop shows the back of a customer's head, using a second mirror. 5

OR

Describe how the position of sunrise and sunset shifts along the horizon through the year and explain using the tilt. 5

END OF QUESTION PAPER Total: 50 Marks

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